

Admitted Student Frequently Asked Questions

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Program Practicalities

Q: What is the duration of the program? Can I complete the program in one year?

A: The MSE (often referred to as MSE "Professional"), MSE-SS and MSE-ES programs are 16 months long, on-campus, and require students to be full-time. It is impossible to complete any of the programs in less time than the stated duration.

Q: What is the anticipated class size?

A: The estimated combined class size for MSE is 25-30 students, and for MSE-SS and MSE-ES is 50-55 students total.

Q: What is the professor-student ratio?

A: The professor-student ratio is currently 1:5.

**Q: What are the tuition and other charges for the 2025-2026 academic year (AY)?
What is the schedule of payment?**

A: To give you an idea of what to expect, the MSE-SS and -ES tuition rate for the 2025-26 academic year will be \$31,500.00 per semester, and the MSE tuition rate will be \$26,250.00 per semester. You can also find the same tuition rates listed on the [website](#) and you may visit the [2025-26 University Fees page](#) for a list of other charges. Information about the schedule of payment can be found on the [CMU Student Financial Services website](#).

Q: What is the approximate I-20 amount for the MSE-SS program?

A: Please review OIE's pre-arrival FAQ page that can assist with many questions that new students have: <https://www.cmu.edu/oie/faq/pre-arrival-new-student.html>. The question for the cost of living is categorized as an "Estimated Expense." You can find a breakdown of your estimated expense total on this page as "What is the total amount I need to show on my financial documents?"

Q: Is part-time CPT allowed along with any semester, except the mandatory summer internship?

A: No

Q: How many hours per week are required for each mini course and for the Studio Project?

A: Mini courses are 6 units. Studio Project unit counts vary by semester but range from 12-36 units. Each unit typically corresponds to one hour per week, so a 12-unit course should take approximately 12 hours of work per week (spent in class lectures, readings, individual assignments, group projects, etc.).

Q: Are the MSE programs a part of CMU's School of Computer Science?

A: Yes, all of the MSE programs are housed in the Software and Societal Systems Department (S3D), which is a department within the School of Computer Science . For a list of the seven departments in SCS, along with each of their master's programs, please visit this [page](#).

Curriculum

Q: How similar / different are the MSE-ES and -SS tracks?

A: Scalable Systems students study large-scale, intelligent systems and take the following courses:

- 17-626 Requirements for Information Systems
- 17-647 Data-intensive and Scalable Systems
- 17-695 Design Patterns

Whereas the Embedded systems students study topics such as Internet-of-Things and cyber physical systems and take these courses:

- 17-627 Requirements for Embedded Systems
- 17-648 Sensor-based Systems
- 17-638 Engineering Embedded Systems

Otherwise, the two tracks have a similar focus on first principles and foundations. Students can further distinguish their track by selecting electives that fit within their specialization.

Q: Based on what I heard, it seems students from MSE Professional and MSE-SS/ES attend the same courses? How will the courses be designed, given that they are two very different groups of engineers and probably have different levels of knowledge/skills?

A: When appropriate, there are two different sections of a course, one for MSE for professional students and another for MSE-SS/ES students. There are also some curricular differences between the programs.

Q: Is the MSE Professional program more inclined towards preparing students for leadership or managerial roles in the software industry?

A: Students can expect that they will be equipped to take on almost any role upon graduation. However career outcomes are influenced by a combination of factors such as student background and skills acquired in the program, etc.

Q: What courses are recommended, if any, for students to take in the summer prior to matriculation?

A: Students are not required to take any courses in the summer prior to starting the program. An optional, 6-unit course, 15-503 Introduction to Computer Systems, is available. This course is often a prerequisite for other courses in the School of Computer Science so some students opt to take it before they arrive. It is important to note that 15-503 is not a required pre-req for any MSE courses.

Q: Are there any restrictions on available electives? Can we take electives from other departments within the School of Computer Science (SCS)?

A: Students will typically take electives within the Software and Scalable Systems Department (S3D) but ultimately students may choose to take electives outside of our department as long as they (1) have not completed the course before, and (2), they have consulted with the Senior Graduate Academic Advisor, Lauren Martinko, and received approval to register for that course.

Q: Are there any prerequisites that we need before joining the program?

A: No.

Q: I would like to take more than the required number of elective courses outlined in the Plan of Study. Can I do that?

A: In the first semester of the program, you may *not* take more than the required number of units as outlined in the Plan of Study. There are no exceptions. CMU's coursework is extremely rigorous, and the recommended number of units per semester is 51, well above the full-time unit requirement of 36.

Q: Can we take elective courses in the first fall semester?

A: MSE students take 12 units of electives in the first fall semester. MSE-SS/ES students do not have the option of taking an elective in the first fall semester.

Q: Can I take *any* CMU course, or are there some restrictions?

A: There are some restrictions. In general, students are permitted to take courses outside of the Software and Societal Systems Department (S3D) but they should be applicable to your software engineering degree. For example, a student may not be able to take a course on piano and expect to have it count towards degree completion. We understand that some students do want to explore outside interests and there may be other options available including taking the course as an audit or pass/no pass. Elective choices must be made in consultation with the Senior Graduate Academic Advisor, Lauren Martinko, prior to registration. Lauren will know which courses can be applied to your degree program, and which ones will not.

Q: Is it possible to talk to the graduate academic advisor prior to the start of the program?

A: Lauren Martinko, the Senior Graduate Academic Advisor, will reach out to you prior to the start of the program with information about the plan of study for your program and instructions on how to register for classes.

Q: Am I required to take 15-503 (previously 15-513)?

A: **Taking 15-503 is optional.** None of the courses in the MSE program require you to take 15-503. There are, however, a number of electives outside the MSE program that do list 15-503 as a prerequisite.

Q: Which courses list 15-503 as a prerequisite?

A: Electives commonly taken by our students for which 15-503 is a prerequisite are: 15-619 Cloud Computing; 15-640 Distributed Systems; and 15-719 Advanced Cloud Computing. Please note that this is not a complete list of courses for which 15-503 is a prerequisite.

Q: What is the process for enrolling in 15-503 over the summer if I want to take electives that list 15-503 as a prerequisite?

A: You will be asked whether or not you plan to enroll in 15-503 on your enrollment form in your application portal. When deciding to enroll, please consider the following:

- The course is very time intensive, so we often caution against taking it over the summer for anyone who has other obligations like work or school.
- Note that if this course is taken in the summer before the start of the MSE program, it will not be counted toward degree completion, but the earned grade will be factored into your cumulative QPA.
- The cost for the course will be approximately \$3100. The summer semester payment deadline is May 15th. Student Financial Services handles all of the billing and payments for CMU students. An invoice will be posted in Student Information Online (SIO). Please visit their [website](#) if you have any questions.

Q: Is there a tentative date of registrations opening for 15-503? Is there a limit on the number of seats available in the class?

A: While registration is currently open, there is no urgency to register at this moment since there are an unlimited number of "seats" in the course because it is offered remotely. Students have until mid-May to register.

Q: Do any of the courses go into software documentation?

A: Software is documented in many ways: requirements, such as user stories, architectural diagrams, and comments in source code. While this topic is not a principal topic of any course, it should be covered as a consequence of learning and practicing these activities.

Q: Is there an option to show competency in a certain core course and replace them with electives?

A: No. All core, required and project courses must be completed through Carnegie Mellon University's Master of Software Engineering program. Students may petition for credits to apply toward their elective unit requirement if the course transfer conditions are met. The details of this process may be found on pages 35 and 36 of the [MSE Handbook](#).

Q: When is it advisable (which semester) to take flagship SCS courses like Intro to ML / DL, Cloud Computing ,etc ?

A: It depends on whenever they are being offered, if you can get a seat, and when you have room in your schedule. You can find the [schedule of classes](#) on the CMU website.

Capstone Projects

Q: What are the capstone projects and how do they differ?

A: The Studio Project is for MSE Professional students, whereas the Practicum Project is for MSE-SS and MSE-ES students. The Studio Project and Practicum Project differ in duration and scope. Studio begins in the Spring semester and continues through the Summer and Fall whereas Practicum begins and ends in the 2nd Fall semester.

Q: Can an MSE student complete an internship instead of the MSE Studio Project?

A: No. The Studio Project has a much larger scope, and a longer duration than a typical internship.

Q: In the course curriculum, it's written that Studio Projects are mostly sponsored by external clients - does this mean financial sponsorship to students as well since we build solutions to their problems/pain points?

A: It does not. You will receive course credit. Each project team is assigned 2 mentors and Practice Area Coaches that guide and advise students throughout the entire duration of the project. The money received from sponsors goes directly towards the compensation of the project mentors and coaches.

Q: What will be evaluated for the final results?

A: Assessment consists of project initiation, practice area evaluations, project outcomes, stakeholder communication, and individual growth. You can learn more about the assessment expectations by reading the [17-671 Studio Project syllabus](#).

Q: Are students expected to purchase needed software and components, like cloud virtual instances if required for the project?

A: All necessary software and equipment will be provided at no cost to the student.

Q: Are the capstone projects decided by the program and students pick their preference among them?

A: Program administrators and faculty curate a variety of projects for students to select from. While we try very hard to give every student their first choice of project, there are several other factors considered when forming student teams.

Q: Is it suggested to take heavy elective courses like 15-619 Cloud Computing along with the studio projects?

A: Your academic advisor will be able to help you determine if you have the ability and capacity to take a demanding elective while engaged in the studio project.

Q: Have students continued with their project/company after completion of the capstone projects?

A: There have been a handful of students who have gone on to work with the company that sponsored the capstone project they worked on.

Q: Could you highlight the benefits of doing the Studio Project over doing an internship?

A: Internship is a part of the ES/SS students' curriculum and provides the necessary industry experience which is especially important since most ES/SS students usually have little-to-no such experience.

Studio is a part of MSE Professional students' curriculum and it allows MSE Professional students, who already have industry experience, to work on a real-world problem. The key objectives are to learn to build a team, work in a group, try-on different project roles and learn client management skills.

Both internship and Studio allow students to establish relationships with the company and potentially receive a full-time offer upon completion.

Q: What do you think is the most challenging thing that students face in the capstone projects?

A: An often repeated response in the MSE Programs is, "it depends." This is no exception. There are many challenges that students face, but if pressed to pick one, it would be time management.

MSE-SS/MSE-ES Internships

Q: How difficult is it to get an internship?

A: The MSE program has maintained a 100% placement for internships. There are a lot of resources available to help with the search process. Although we provide staff and faculty support throughout the internship search process, securing an internship, whether on or off campus, is ultimately the student's responsibility.

Q: Can we extend graduation if we did not find an internship?

A: No.

Q: Do you have career fairs? To what extent does the program help in finding internships?

A: There are several career fairs throughout the Spring and Fall semesters offered by CMU's Career and Professional Development Center. The program and Career Services provide opportunities for students to attend tech talks, career fairs, share resumes, and connect with alumni at companies.

Q: Are there any restrictions on the kind of internship job profiles that we can accept?

A: Yes, internships need to be software engineering focused and related to the student's program track if possible.

Q: Would there be any on-campus hiring events such as career fairs? How would the university support students in their search for an internship?

A: There are many resources available to help with the search. There are career fairs in the Spring and Fall semesters along with tech talks hosted by the [Career and Professional Development Center](#) (CPDC) and the MSE program too.

Q: Is the summer internship counted toward CPT/OPT? Are there sources for international students about typical visa status and requirements during and after the program?

A: The [Office of International Education \(OIE\)](#) supports international students throughout the visa process, including the process for OPT/CPT. Summer internship is counted towards CPT for international students only. For more information about the CPT/OPT process, please refer to the [OIE webpage](#).

Q: Do most students intern at their top-choice company?

A: Many factors go into landing an internship including market forces, location concerns, salary, etc. While most students do get the opportunity to work at their top choice or similar, we recommend that students keep an open mind and consider all options.

Q: Are there any restrictions on the kind of internship job we can accept?

A: Yes, internships need to be relevant to software engineering and your program and meet certain learning objectives and time requirements.

Q: Can students take up a part-time internship during the academic year?

A: No. Internships cannot be completed outside of the summer semester. CPT can only be approved with a required internship course, which takes place in the summer for the MSE-SS/ES programs.

Q: Do the domains of the internships/full-time jobs usually align with the ES/SS specializations done? Or are the types of jobs similar for both?

A: Internships tend to be more generalized, but there are opportunities to target specific internships that have a focus on embedded systems or scalable systems. For example, MSE-ES students may target internships with companies that do work on embedded systems such as Uber or robotics companies. For full-time positions, there are opportunities to be more focused (Infrastructure Engineer, Software Developer-Cloud, etc). A lot of times "Software Developer" is the title, but the job description will be more detailed on the work that position will be doing and can be reflective of both tracks. We see similar titles received by graduates across all programs.

Q: What would be the ideal time for me to start my internship search?

A: Most SS/ES students begin their search immediately upon entering the program in September. The search for an internship may continue into late spring.

MSE Professional Program Internships

Q: Are MSE students able to do summer internships?

A: MSE students are not eligible for internships and will be enrolled in Studio Project over the summer, which is 36 units and equivalent to a full-time job. If you are an international student studying on an F-1 visa, there are also visa implications that prevent MSE students from completing an internship or being approved for CPT.

Career Prospects & Assistance

Q: What is the intended job role for a graduate at the end of the degree? Would it be a Software Developer role or a more senior-level position?

A: It can be either and depends on the individual. Some students decide to go the individual contributor route and some students look to take electives to give them more experience on the managerial route.

Q: What kind of job in which company could previous graduate students enrolled in the program find?

A: Previous graduates have found jobs in various companies ranging in size and focus. The [career destination slides](#) may be helpful in giving you a better idea of where students ended up.

Q: What would be the ideal time for me to start my job search?

A: Late June/early August (third semester) is a good time for students in all programs to begin searching for a job.

Q: Is there any indication that hiring companies prefer a more general degree, such as an MSCS?

A: No. If anything, companies look for MSE students specifically because they have strong skills to be a tech leader throughout the whole software life cycle.

Q: What are the job placement rates for the MSE programs class of 2024?

A: We are still collecting data on the class of 2024 since they just graduated in December. In the meantime, the [career destination slides](#) from the class of 2023 should give you a general idea.

Q: What types of companies typically visit the career fairs hosted by CMU?

A: You can find more info on career fairs here:

<https://www.cmu.edu/career/employers/calendar.html>

Financial Aid/Scholarships

Q: Please provide information on funding sources.

A: The majority of students fund their education through loans but other options may be available.

Students who are U.S. Citizens or permanent residents may be eligible for Federal Financial Aid. More information about this can be found [here](#).

[Prodigy Finance](#) is an organization that provides loans to international students earning degrees in STEM fields, so this may be worth checking out.

Additionally, the Carnegie Mellon Financial Services [webpage](#) refers to FASTChoice. [FASTChoice](#) is a loan comparison service that may be able to provide private loan options for you.

Please note: Carnegie Mellon does not endorse any lender listed within FASTChoice. Students and families have the right to choose any lender for private loans. Carnegie Mellon does not accept any revenue sharing funds from any lender.

Q: Please provide more details on the scholarships listed in the program's website, specifically the amount and qualification criteria. Are these open to international students too?

A: The MSE programs offer two scholarships: (1) two James E. Tomayko Scholarships awarded for academic and leadership excellence, typically in the final program semester for \$7,500 each; (2) five Director's Scholarships awarded for academic and leadership excellence for \$7,500 each, typically awarded to students' final program semester.

All students are eligible for merit scholarships awarded by the program. More information about scholarships can be found on the [website](#).

Working

Q: What types of Teaching Assistantships (TA) are available? How many such positions are available? What are the criteria for getting selected for one? From which semester onwards do you allow students to TA in courses? What are the requirements for attaining a TA-ship?

A: MSE Teaching assistantships are generally available in the Spring and second Fall semesters. MSE has approximately twenty teaching assistantships that are assigned based on student eligibility, good academic standing, and experience with the course topic. Teaching assistantships pay approximately \$18/hour. This amount may vary by department or program. International students are limited to a maximum of 20 hours/week. TAs are paid on a bi-weekly basis.

Q: Is it possible to TA undergraduate level courses, or only courses within the MSE program?

A: Graduate students do serve as TAs for undergraduate courses. You may, however, find it difficult to obtain positions outside the MSE as you will be competing with graduate students in those departments. For example, a calculus class will often rely upon graduate students in the Mathematical Sciences program to serve as TAs.

Q: What about RA positions?

A: Students interested in research typically register for an independent study with a faculty advisor first, before being considered for a paid research assistantship (RA). RA positions are not a principal part of the professional masters degree programs.

Q: Apart from being a TA/RA, can MSE students have on-campus jobs? What opportunities are available to graduate students?

A: International students are not eligible to work while on an F-1 visa. U.S. Citizens and permanent residents may work while they are enrolled, but it's not encouraged while taking courses full-time, with the exception of being a TA. Check out this [website](#) for more information about potential on-campus work opportunities.

Q: How do I find a TA/RA position prior to arriving in Pittsburgh?

A: First semester students are strongly discouraged from holding a job while they are acclimating to Pittsburgh, CMU and the MSE program.

Miscellaneous

Q: How many students usually continue after the program to do a Ph.D. as opposed to a job in industry? Is it possible to further pursue the Software Engineering Ph.D. Program in S3D after finishing the MSE degree program?

A: A very small number of MSE graduates go on to pursue a Ph.D. And, yes, it is possible to pursue a Ph.D. in Software Engineering in S3D upon completion of the Master's degree.

Q: What research opportunities are available to MSE program students?

A: The MSE programs are professional programs aimed at supporting students focused on finding full time employment as software engineers. Students may engage in research on campus through independent studies, if approved by at least one faculty member and the Senior Graduate Academic Advisor, to count toward elective units for degree requirements.

Q: I'm interested in embedded software with regards to robotics. Does the MSE program have any resources or opportunities that would enhance my software development skills and also support my interest in robotics?

A: The MSE program for professionals allows students to take up to four free electives across CMU, which is intended to help students specialize in specific areas, such as Robotics or Machine Learning. Students who want to specialize in Robotics, for example, should consider using electives to enroll in graduate-level Robotics courses, and could contact a Robotics faculty member to express interest in their research and in an independent study as a means to engage with the researcher and their lab. This request is typically sent by the student to the faculty member after completing one of the faculty member's courses, in which case the faculty member has some familiarity with the student, and the student has familiarity with core concepts used in the lab, before the faculty member agrees to supervise the study.

Q: Can admitted students visit campus and meet with professors, program directors, advisors, and current students?

A: If you would like to travel to Pittsburgh to get a better feel for the city, the CMU campus, and the MSE space, you are welcome to organize your own visit. You may contact your Admissions Officer, Marlana Ivey, prior to your visit to see if additional accommodations can be made. Due to the rigid schedule of the program, though, meeting with individual members of the MSE community is often limited or not possible. To give you a feel for the MSE space in which students study and attend class, we filmed a [tour](#).

Q: What kind of laptop do you recommend for incoming students?

A: Here are some recommendations from SCS Computing Facilities:

<https://computing.cs.cmu.edu/desktop/recom-configs>

Q: What's a good time for international students to arrive in Pittsburgh?

A: Students typically arrive in Pittsburgh at the end of July through mid-August. The Office of International Education will provide all incoming international students with details on the date of earliest arrival as well as the latest arrival date that is legally permitted. Find more info under the "Arrival in the U.S." section:

<https://www.cmu.edu/oie/faq/pre-arrival-new-student.html>

Q: What other options do we have if there is a 1-2 week delay in arrival after the start of classes? Can we take online classes for these 2 weeks?

A: Students are required to be here at the start of classes so options are limited if someone arrives later and may be faced with denied entry into the United States. Therefore, students should all plan to arrive before the start of class. If an emergency or unforeseen circumstance occurs, students need to contact your Admissions Officer, Marlana Ivey, as soon as possible. She, along with the help of your OIE Advisor, will assist accordingly.

Q: Are there housing facilities available on Campus?

A: Campus housing is only available to undergraduate students.

Q: Could you share a few links to search for apartments?

A: [zillow.com](https://www.zillow.com)
[apartments.com](https://www.apartments.com)
[trulia.com](https://www.trulia.com)

Many graduate students use unofficial Facebook groups to find roommates, sublets, and leases near CMU. Some active and reliable groups are listed below.

[Carnegie Mellon University Housing, Roommates, Tenants, Apartments](#)

[Carnegie Mellon University \(CMU\) Apartments and Housing](#)

[Carnegie Mellon University \(CMU\) Off Campus Housing & Roommates](#)

[Carnegie Mellon University \(CMU\) Housing, Sublets & Roommates](#)

[Carnegie Mellon University \(CMU\) Housing, Sublets & Roommates Group 2 Renting/Subletting @ CMU](#)

[Carnegie Mellon University \(CMU\) Housing, Sublets & Roommates Group 2](#)

[University of Pittsburgh \(PITT\) Housing, Sublets & Roommates](#)

[University of Pittsburgh \(PITT\) Housing, Sublets & Roommates](#)

[Pitt and CMU Housing, Sublets, and Roommates](#)

Please note- these are not official university-sanctioned groups. The university does not partner with any resources listed above.